TROUBLESHOOTING GUIDE

Dredging Equipment — All Series

Marine Engineering Division

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Prepared for: Benamdaj Dredging Solutions Ltd.

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1. Introduction & How to Use This Guide

This Troubleshooting Guide covers all four Benamdaj dredger models: the 16/14" Cutter Suction Dredger (Weichai 6170 engine, marine gearbox), the 14/12" Amphibious Multifunctional Dredger (Weichai 6170, marine gearbox), the 12/10" Bucket Chain Dredger (CAT 3408, marine gearbox), and the 10/10" Jet Suction Dredger (Weichai WP13, PTO drive). All models use a CAT 3412 auxiliary engine for electrical generation.

Procedures in this guide are largely identical across models since all four share the same winch-based manual positioning, the same electrical system, and the same high-chrome centrifugal slurry pump type. Where a procedure differs by model this is explicitly noted.

HOW TO USE: (1) Find the symptom in the Quick Reference index at Section 2. (2) Go to the relevant fault block. (3) Follow the diagnostic steps in order. (4) Apply the fix. (5) Apply the prevention steps to avoid recurrence.

1.1 Fault Escalation Levels

Level	Who Responds	Examples
Level 1 — Operator	Trained vessel operator	Fuel top-up, oil top-up, anchor adjustment, suction blockage, fuse replacement, slurry density adjustment
Level 2 — Senior Technician	Experienced workshop technician	Impeller replacement, seal replacement, gearbox oil change, pump overhaul, belt replacement, wiring repair
Level 3 — Specialist Engineer	Factory or OEM engineer	Engine internal failure, gearbox rebuild, structural crack repair, shaft replacement, major electrical fault

2. Quick Symptom Reference Index

SYMPTOM	MOST LIKELY FIRST ACTION
Engine will not start	Check fuel, battery, oil level — see Section 3.1
Engine starts then stalls	Check fuel filter, air filter, idle speed — see Section 3.2
Engine overheats	Stop immediately. Check coolant fan, check for blockage — see Section 3.3
Low oil pressure warning	Stop immediately. Check oil level, check for leak — see Section 3.4
Engine loses power under load	Check air filter, fuel filter, gearbox engagement — see Section 3.5
Black smoke from exhaust	Reduce load, check air filter — see Section 3.6
White/blue smoke from exhaust	Check for oil or water in combustion — see Section 3.7
Engine knocking or rattling	Stop immediately. Low oil or bearing failure — see Section 3.8
High fuel consumption	Check air filter, injectors, load — see Section 3.9
Gearbox overheating	Check oil level, reduce load — see Section 3.10
Gearbox oil leak	Identify seal, stop and repair — see Section 3.11
Pump loses suction / cavitates	Check suction depth, check for blockage, check pressure — see Section 4.1
Pump output drops suddenly	Check suction blockage, check impeller wear — see Section 4.2
No flow from discharge pipe	Check for blockage, check pump priming — see Section 4.3
Pump noise — banging or grinding	Stop immediately. Debris or impeller damage — see Section 4.4
Pump seal leaking	Check gland packing, tighten or replace — see Section 4.5
Suction pipe blocked	Raise arm, clear obstruction, restart — see Section 4.6
Discharge pipe blocked	Stop pump, locate and clear blockage — see Section 4.7
Winch will not hold tension	Check brake, check wire condition — see Section 5.1
Anchor wire snaps	Stop operations, reset anchor — see Section 5.2
Dredger swings off course	Check anchor tensions, adjust winch — see Section 5.3
Solar / generator not working	Check connections, check fuse — see Section 6.1
Battery flat / not charging	Check alternator, check terminals — see Section 6.2
Production rate low	Trace cause: pump, engine, soil type, depth — see Section 6.3
Discharge distance reduced	Check pump wear, check pipe blockage — see Section 6.4
Slurry too watery	Increase suction depth, reposition anchor — see Section 6.5

3. Engine & Gearbox Faults

The following fault blocks apply to all models. Note that Models 1, 2, 3 use a marine gearbox between engine and pump. Model 4 uses a PTO (Power Take-Off) instead — gearbox-specific faults do not apply to Model 4.

3.1 ENGINE FAULTS

SYMPTOM: Engine will not start	
Probable causes (most likely first)	1. Insufficient fuel — tank empty or fuel valve closed. 2. Flat or failed battery. 3. Engine oil below minimum level — safety cut-out activated. 4. Clogged fuel filter. 5. Faulty ignition switch.
Diagnostic steps	Check fuel gauge first. Then check battery terminals (clean and tight?). Check oil dipstick. Try cranking — if starter motor sounds weak, battery is low. If starter motor does not engage, check ignition fuse.
Step-by-step fix / repair	1. Refuel if empty. 2. Charge or replace battery. 3. Top up engine oil to correct level. 4. Replace fuel filter. 5. Check ignition fuse and replace if blown. 6. If none of the above resolves it, call Level 2 technician.
Parts / materials required	Fuel filter, engine oil (Total Ruby X), battery, ignition fuse
Tools required	Spanners, voltmeter, funnel
Estimated repair time	15–45 minutes for operator-level fixes
Operator fix or specialist needed?	Level 1 (operator) for fuel, battery, oil. Level 2 for fuel filter, ignition fault.
How to prevent recurrence	Check oil, fuel, and battery at every pre-start checklist. Replace fuel filter at every 200-hour service.
Warranty impact	No — routine consumables are not warranty items.

SYMPTOM: Engine starts then immediately stalls	
Probable causes (most likely first)	1. Fuel starvation — blocked fuel filter or air in fuel line. 2. Idle speed set too low. 3. Engine not warmed up before loading. 4. Water in fuel.
Diagnostic steps	Check if engine runs smoothly at idle without load. If stalls only under load: check gearbox engagement and pump back-pressure. If stalls at idle: check fuel filter and idle adjustment.
Step-by-step fix / repair	1. Replace fuel filter and bleed fuel system. 2. Adjust idle speed to manufacturer's minimum (confirm with engine manual). 3. Drain water from fuel separator. 4. If engine runs without gearbox engaged but stalls when pump is loaded — check pump for mechanical blockage.
Parts / materials required	Fuel filter, fuel separator element
Tools required	Spanners, screwdriver, bleed kit
Estimated repair time	30–60 minutes
Operator fix or specialist needed?	Level 1 for idle and fuel. Level 2 for injector or pump issues.
How to prevent recurrence	Replace fuel filter every 200 hours. Use clean, dry diesel fuel storage. Drain water separator weekly.
Warranty impact	No — fuel system maintenance is operator responsibility.

SYMPTOM: Engine overheats — temperature alarm at 105°C

Probable causes (most likely first)	1. Cooling fan blocked or not operating. 2. Engine overloaded — running at excessive RPM for extended period. 3. Low engine oil causing increased friction heat. 4. Blocked air intake. 5. Engine room restricted airflow.
Diagnostic steps	Stop engine immediately when alarm sounds. Allow to cool 10 minutes. Check cooling fan rotation. Check oil level. Check air intake for blockage. Check engine compartment ventilation.
Step-by-step fix / repair	1. Clear any blockage from cooling fan and air intake. 2. Check and top up engine oil. 3. Reduce engine speed — do not run at maximum RPM for extended periods in hot ambient conditions. 4. Ensure engine compartment has adequate ventilation. 5. If overheating recurs, call Level 2 technician.
Parts / materials required	Engine oil (Total Ruby X)
Tools required	Spanners
Estimated repair time	Cool-down 10–15 minutes. Fix 15–30 minutes.
Operator fix or specialist needed?	Level 1 for oil top-up, blockage clearing. Level 2 for fan motor failure.
How to prevent recurrence	Check oil level and air intake every day. Maintain engine compartment ventilation. Do not run engine above rated RPM for extended periods.
Warranty impact	Depends on cause — overloading or lack of oil maintenance voids warranty.

SYMPTOM: Low engine oil pressure warning

Probable causes (most likely first)	1. Engine oil level below minimum. 2. Engine oil pump failure. 3. Blocked oil filter. 4. Oil leak — external or internal. 5. Oil viscosity incorrect for ambient temperature.
Diagnostic steps	STOP ENGINE IMMEDIATELY. Low oil pressure causes rapid bearing failure. Check oil level on dipstick. Inspect underside of engine for oil leak. Check oil filter condition.
Step-by-step fix / repair	1. Top up engine oil to MAX mark and recheck pressure. 2. Replace oil filter at next service if blocked. 3. If external oil leak is visible, locate and seal source. 4. If oil level is correct but pressure is still low — call Level 3 specialist (internal oil pump or bearing failure).
Parts / materials required	Engine oil (Total Ruby X), oil filter
Tools required	Spanners, oil drain tray
Estimated repair time	Oil top-up: 10 minutes. Seal leak: 1–3 hours. Internal failure: specialist required.
Operator fix or specialist needed?	Level 1 for oil top-up. Level 3 for internal pressure failure.
How to prevent recurrence	Check oil level every day before startup. Change oil and filter every 200 hours.
Warranty impact	Failure due to running without oil voids warranty.

SYMPTOM: Engine loses power under load

Probable causes (most likely first)	1. Clogged air filter restricting combustion air. 2. Clogged fuel filter restricting fuel supply. 3. Pump back-pressure too high — slurry too dense or discharge blocked. 4. Engine RPM not set to rated operating speed. 5. Engine operating above its rated continuous duty power.
Diagnostic steps	Check air filter — hold up to light, replace if blocked. Check fuel filter. Check pump pressure gauge — if above 10 bar, reduce RPM and investigate discharge line. Confirm engine RPM is within 1,200–2,100 range.

Step-by-step fix / repair	1. Replace air filter. 2. Replace fuel filter. 3. Clear discharge pipe if blocked. 4. Reduce pump load — raise suction arm slightly to reduce density. 5. Allow engine cool-down if thermal derate is suspected.
Parts / materials required	Air filter, fuel filter
Tools required	Spanners
Estimated repair time	Filter replacement: 20–30 minutes
Operator fix or specialist needed?	Level 1 for filters and pump adjustment. Level 2 for fuel injection issues.
How to prevent recurrence	Replace air and fuel filters every 200 hours. Monitor pump pressure continuously.
Warranty impact	No — filter maintenance is operator responsibility.

SYMPTOM: Excessive black smoke from exhaust

Probable causes (most likely first)	1. Clogged air filter — insufficient combustion air causing rich mixture. 2. Engine severely overloaded. 3. Injector fault — fuel not atomising correctly. 4. Engine running in very high ambient temperature.
Diagnostic steps	Check air filter first — this is the most common cause. If filter is clean, check engine load — reduce RPM and see if smoke clears. If smoke continues at low load, call Level 2 for injector check.
Step-by-step fix / repair	1. Replace air filter immediately. 2. Reduce engine load. 3. Level 2: injector cleaning or replacement.
Parts / materials required	Air filter, injector set (Level 2)
Tools required	Spanners, injector tool (Level 2)
Estimated repair time	Air filter: 20 minutes. Injectors: 2–4 hours (Level 2)
Operator fix or specialist needed?	Level 1 for air filter. Level 2 for injectors.
How to prevent recurrence	Replace air filter every 200 hours. Inspect weekly in dusty/sandy environments.
Warranty impact	Injector failure: may be covered under warranty if within service life.

SYMPTOM: Unusual knocking or rattling from engine

Probable causes (most likely first)	1. Low engine oil — bearing knock. 2. Loose engine mounting bolts. 3. Damaged impeller contacting pump casing (transmitted vibration). 4. Faulty injector causing detonation knock. 5. Loose exhaust manifold.
Diagnostic steps	STOP ENGINE if knocking is severe. Check oil level. Check engine mounting bolts. Listen to isolate source — is noise from engine, gearbox, or pump? Remove pump drive to determine if engine or pump is source.
Step-by-step fix / repair	1. Top up oil and recheck. 2. Tighten engine mounting bolts to correct torque. 3. If pump-source: inspect impeller for damage. 4. Level 2/3: internal engine knock requires specialist.
Parts / materials required	Engine oil, bolts, impeller (if pump source)
Tools required	Torque wrench, spanners
Estimated repair time	Oil and bolts: 30 minutes. Internal engine: specialist required.
Operator fix or specialist needed?	Level 1 for oil and bolts. Level 3 for internal knock.
How to prevent recurrence	Check oil daily. Check mounting bolts monthly. Replace impeller on schedule.

Warranty impact	Internal knock due to oil starvation voids warranty.

3.2 MARINE GEARBOX FAULTS (MODELS 1, 2, 3 ONLY — NOT APPLICABLE TO MODEL 4)

SYMPTOM: Gearbox overheating	
Probable causes (most likely first)	1. Low gearbox oil level. 2. Wrong grade of gearbox oil. 3. Gearbox overloaded — pump back-pressure too high. 4. Prolonged operation without adequate cool-down.
Diagnostic steps	Check gearbox oil level and condition every morning. If gearbox is hot to touch after normal operation: check oil level, check pump pressure for overload.
Step-by-step fix / repair	1. Top up gearbox oil to correct level with correct grade. 2. Allow cool-down. 3. Reduce pump back-pressure — check for discharge blockage. 4. If overheating persists: Level 2 inspection of gearbox internals.
Parts / materials required	Gearbox oil (confirm grade with manufacturer)
Tools required	Spanners, oil drain tray
Estimated repair time	Oil top-up: 15 minutes. Cool-down: 20–30 minutes.
Operator fix or specialist needed?	Level 1 for oil and cool-down. Level 2 for persistent overheating.
How to prevent recurrence	Check gearbox oil every morning. Change every 200 hours. Never exceed pump pressure limits.
Warranty impact	Damage from running with low oil or wrong oil grade voids warranty.

SYMPTOM: Gearbox oil leak	
Probable causes (most likely first)	1. Worn or damaged shaft seal. 2. Loose drain plug or inspection cover. 3. Cracked gearbox casing from impact.
Diagnostic steps	Identify leak location — is it from shaft seal, drain plug, or casing? Check oil level to determine severity of loss.
Step-by-step fix / repair	1. Tighten drain plug or inspection cover. 2. Level 2: replace shaft seal — requires gearbox dismantling. 3. Level 3: casing crack requires specialist or replacement.
Parts / materials required	Shaft seal kit, gasket set, drain plug washer
Tools required	Spanners, seal driver, torque wrench
Estimated repair time	Drain plug: 15 minutes. Seal replacement: 2–4 hours (Level 2).
Operator fix or specialist needed?	Level 1 for drain plug. Level 2 for seal. Level 3 for casing.
How to prevent recurrence	Check gearbox for leaks every morning. Replace shaft seal at major overhaul (90-day service).
Warranty impact	Seal wear is normal wear — not typically warranty. Impact damage may be assessed case by case.

4. Pump & Suction System Faults

The pump is the most critical component for dredging output. Reduced pump performance directly reduces production. Monitor pump pressure (normal: 3 bar, max: 10 bar) continuously during operation.

SYMPTOM: Pump loses suction / cavitates	
Probable causes (most likely first)	1. Suction head not sufficiently submerged — pump drawing air. 2. Air ingress through suction pipe joint or fitting. 3. Slurry density too high — pump cannot maintain suction. 4. Suction pipe partially blocked. 5. Worn impeller reducing suction efficiency.
Diagnostic steps	Signs of cavitation: pressure gauge drops, pump output reduces, loud rattling noise from pump. Check: is suction head submerged? Check suction pipe fittings for air ingress. Check pressure reading.
Step-by-step fix / repair	1. Lower suction arm until head is fully submerged. 2. Check and tighten all suction pipe joints. 3. Reduce pump RPM to reduce slurry density entering pump. 4. Adjust system pressure to maintain pressure above cavitation threshold. 5. If impeller worn — Level 2 replacement.
Parts / materials required	Suction pipe fittings, pipe clamps, impeller (if worn)
Tools required	Spanners, pressure gauge
Estimated repair time	Repositioning: 10–15 minutes. Seal: 30 minutes. Impeller: 4–8 hours (Level 2).
Operator fix or specialist needed?	Level 1 for depth and fittings. Level 2 for impeller replacement.
How to prevent recurrence	Monitor pump pressure continuously. Replace impeller every 200 hours or when output reduces noticeably.
Warranty impact	Cavitation damage to impeller from running dry or incorrect operation may void warranty.

SYMPTOM: Pump output drops suddenly during dredging	
Probable causes (most likely first)	1. Suction pipe blockage — large object or debris. 2. Discharge pipe blockage. 3. Worn impeller — gradual loss of efficiency. 4. Engine RPM dropped — check engine. 5. Suction head lifted off dredging face.
Diagnostic steps	Check pump pressure — if pressure drops: suction blockage. If pressure rises: discharge blockage. Check engine RPM. Check suction head position.
Step-by-step fix / repair	1. Suction blockage: raise arm from obstruction using winch, lower back and restart. 2. Discharge blockage: stop pump, loosen pipe joints to locate and clear blockage. 3. Worn impeller: Level 2 replacement. 4. Restore engine RPM.
Parts / materials required	Impeller, pipe couplings
Tools required	Spanners, pipe wrench
Estimated repair time	Repositioning: 15 minutes. Pipe clearance: 30–60 minutes. Impeller: Level 2.
Operator fix or specialist needed?	Level 1 for repositioning and basic blockage. Level 2 for impeller.
How to prevent recurrence	Monitor pressure continuously. Replace impeller on schedule. Inspect discharge pipe joints weekly.
Warranty impact	Normal wear items — not warranty covered.

SYMPTOM: No flow from discharge pipe	
Probable causes (most likely first)	1. Pump not primed — air locked in suction line. 2. Complete discharge pipe blockage. 3. Pump drive not engaged — gearbox or PTO not in drive. 4. Suction head not submerged.
Diagnostic steps	Confirm pump drive is engaged and engine is running. Check suction head is submerged. Check discharge pipe for blockage starting at pump outlet.
Step-by-step fix / repair	1. Prime pump: ensure suction is submerged and restart pump at slow speed. 2. Clear discharge: loosen pipe couplings to locate blockage, flush with clean water. 3. Engage gearbox or PTO correctly.
Parts / materials required	Pipe couplings
Tools required	Spanners, pipe wrench
Estimated repair time	Priming: 10–15 minutes. Discharge blockage: 30–90 minutes.
Operator fix or specialist needed?	Level 1.
How to prevent recurrence	Always prime pump before starting. Flush discharge pipe with clean water at end of every shift.
Warranty impact	N/A — operational issue.

SYMPTOM: Pump seal / gland leaking — slurry from gland	
Probable causes (most likely first)	1. Worn gland packing — normal wear after extended operation. 2. Excessive suction pressure forcing slurry through gland. 3. Shaft scored or worn at seal location.
Diagnostic steps	Inspect gland area during operation. A slow drip (2–3 drops per minute) is acceptable. Steady flow or slurry spray indicates immediate attention needed.
Step-by-step fix / repair	1. Tighten gland follower nuts gradually — do not overtighten. 2. If still leaking: replace gland packing (Level 2). 3. If shaft is scored: Level 3 shaft repair or replacement.
Parts / materials required	Gland packing rope (confirm size with pump manufacturer)
Tools required	Spanners, gland follower tool
Estimated repair time	Gland tightening: 15 minutes. Packing replacement: 2–4 hours (Level 2).
Operator fix or specialist needed?	Level 1 for tightening. Level 2 for packing replacement.
How to prevent recurrence	Inspect gland at every shift. Tighten gland follower regularly. Replace packing at 200-hour service.
Warranty impact	Normal wear — not warranty covered.

SYMPTOM: Suction pipe blocked	
Probable causes (most likely first)	1. Large object (rock, debris, root) lodged in suction pipe or at suction head. 2. Solidified sediment if pump was shut down without flushing.
Diagnostic steps	Pump pressure drops to zero with engine running. No discharge flow. Sound of pump running empty.
Step-by-step fix / repair	1. Raise suction arm away from dredging face using manual winch. 2. If blockage is at head: reposition and restart. 3. If blockage is in pipe: loosen couplings to locate and remove obstruction manually. 4. Flush with clean water before restarting.
Parts / materials required	Pipe couplings

Tools required	Spanners, pipe wrench
Estimated repair time	20–60 minutes
Operator fix or specialist needed?	Level 1.
How to prevent recurrence	Always flush pump and suction pipe with clean water at end of shift. Monitor suction for oversize material.
Warranty impact	N/A.

SYMPTOM: Discharge pipe blocked	
Probable causes (most likely first)	1. Sediment or gravel settled in discharge pipe if pump stopped without flushing. 2. Excessive slurry density causing pipe to plug. 3. Collapsed or kinked flexible section.
Diagnostic steps	Pump pressure rises sharply (above 10 bar) with no discharge flow. Engine may struggle under load.
Step-by-step fix / repair	1. Stop pump immediately to prevent pressure damage. 2. Loosen pipe joints starting from pump outlet, working along pipe. 3. Locate blockage by finding pipe section that does not drain. 4. Clear blockage manually or flush with high-pressure water. 5. Inspect flexible sections for collapse.
Parts / materials required	Pipe couplings, flexible hose section if collapsed
Tools required	Spanners, pipe wrench, high-pressure hose
Estimated repair time	30–120 minutes depending on blockage location
Operator fix or specialist needed?	Level 1 for accessible blockages. Level 2 for buried or submerged pipe sections.
How to prevent recurrence	ALWAYS flush discharge pipe with clean water at end of every shift for minimum 5 minutes. Monitor pump pressure — rising pressure is early warning of discharge restriction.
Warranty impact	N/A — operational issue.

5. Winch, Anchor & Ladder System Faults

SYMPTOM: Winch will not hold tension — anchor line slips	
Probable causes (most likely first)	1. Winch brake worn or not correctly set. 2. Winch drum overloaded — anchor set in wrong position relative to dredger. 3. Wire wrapped incorrectly on drum.
Diagnostic steps	Check brake adjustment — tighten brake if slipping. Check wire spooling on drum for crossovers or tangles.
Step-by-step fix / repair	1. Tighten winch brake adjustment. 2. Reset anchor position closer to vessel. 3. Re-spool wire on drum evenly. 4. If brake mechanism is damaged: Level 2 repair.
Parts / materials required	Winch brake pads/bands if worn
Tools required	Spanners
Estimated repair time	Brake adjustment: 15–30 minutes. Drum re-spool: 30–45 minutes.
Operator fix or specialist needed?	Level 1 for brake adjustment. Level 2 for brake component replacement.
How to prevent recurrence	Inspect winch brakes at every 200-hour service. Never exceed rated winch load.
Warranty impact	Normal wear — not warranty covered.

SYMPTOM: Anchor wire snaps	
Probable causes (most likely first)	1. Wire already damaged — broken strands ignored. 2. Shock load from unexpected current or vessel movement. 3. Wire incorrectly sized for load.
Diagnostic steps	Visual inspection — snapped wire is immediately obvious.
Step-by-step fix / repair	1. Stop all dredging operations immediately. 2. Deploy remaining anchors to hold vessel position. 3. Replace snapped wire with new wire of correct specification before resuming operations.
Parts / materials required	Anchor wire — correct diameter and grade
Tools required	Wire cutters, shackle tool
Estimated repair time	Wire replacement: 30–60 minutes
Operator fix or specialist needed?	Level 1.
How to prevent recurrence	Inspect all anchor wires every morning. Replace any wire showing broken strands or kinks. Never operate with damaged wire.
Warranty impact	N/A — operational consumable.

SYMPTOM: Dredger swings off course during dredging	
Probable causes (most likely first)	1. Unequal tension on port and starboard anchor wires. 2. One anchor dragging — not holding ground. 3. Current acting on vessel hull. 4. Wire fouled on seabed obstruction.
Diagnostic steps	Check tension on all 4 anchor wires — which side has less tension? Is current affecting vessel?
Step-by-step fix / repair	1. Take up slack on the low-tension wire to equalise. 2. If anchor is dragging: stop swing, reposition anchor to firm holding ground. 3. Allow for current direction in anchor placement.

Parts / materials required	N/A
Tools required	Spanners
Estimated repair time	10–30 minutes
Operator fix or specialist needed?	Level 1.
How to prevent recurrence	Set anchors in firm ground before starting swing. Check all tensions before commencing. Account for current in anchor placement.
Warranty impact	N/A.

6. Electrical & Overall Performance Faults

SYMPTOM: Solar power / generator (CAT 3412) will not start	
Probable causes (most likely first)	1. Flat battery — insufficient charge to crank auxiliary engine. 2. Fuel supply to auxiliary engine cut off. 3. Fuse or circuit breaker tripped on generator output circuit. 4. Loose battery terminals.
Diagnostic steps	Check battery voltage. Check fuel supply to CAT 3412. Check all fuses and circuit breakers on generator control panel.
Step-by-step fix / repair	1. Charge or replace battery. 2. Check and restore fuel supply. 3. Reset tripped breakers. 4. Clean and tighten battery terminals. 5. If generator still will not start: Level 2 technician.
Parts / materials required	Battery, fuses
Tools required	Voltmeter, spanners
Estimated repair time	Battery: 30 minutes–2 hours (charging). Fuse: 10 minutes.
Operator fix or specialist needed?	Level 1 for battery and fuse. Level 2 for internal generator fault.
How to prevent recurrence	Check battery daily. Check auxiliary engine fuel level at every pre-start check.
Warranty impact	Generator faults within warranty period — report to Caterpillar dealer.

SYMPTOM: Battery not charging / flat battery	
Probable causes (most likely first)	1. Alternator not charging — belt broken or alternator failed. 2. Battery too old or sulphated. 3. Parasitic drain — equipment left running when engine is off. 4. Corroded terminals preventing charge.
Diagnostic steps	With engine running at operating speed: check voltage across battery terminals — should read 13.8–14.5 V. If less, alternator is not charging. Check alternator drive belt tension.
Step-by-step fix / repair	1. Replace alternator drive belt if broken. 2. Clean battery terminals. 3. Replace battery if old and not holding charge. 4. Check for equipment left on when engine is off.
Parts / materials required	Alternator belt, battery
Tools required	Voltmeter, spanners
Estimated repair time	Belt: 30 minutes. Battery: 30 minutes.
Operator fix or specialist needed?	Level 1 for belt and terminals. Level 2 for alternator replacement.
How to prevent recurrence	Check alternator belt tension weekly. Inspect battery terminals weekly.
Warranty impact	Battery — consumable, not warranty. Alternator — may be warranty if within period.

SYMPTOM: Production rate below expected for soil type	
Probable causes (most likely first)	1. Pump impeller worn — reduced suction and flow. 2. Engine not running at rated RPM — power limitation. 3. Suction head not at optimal depth. 4. Slurry too dilute — mostly water. 5. Discharge pipe restriction reducing flow.
Diagnostic steps	The company's approach: when output is not at maximum capacity, trace the cause back through the system — engine → gearbox/PTO → pump → suction head → discharge pipe. Check each component in turn.

Step-by-step fix / repair	1. Confirm engine is at rated RPM. 2. Check pump pressure — if low, impeller may be worn. 3. Adjust suction depth — deeper position for denser material. 4. Clear any discharge pipe restriction. 5. Reposition anchors to bring suction head to undisturbed material.
Parts / materials required	Impeller (if worn)
Tools required	Pressure gauge, RPM meter
Estimated repair time	Operational adjustments: 15–30 minutes. Impeller: Level 2.
Operator fix or specialist needed?	Level 1 for operational adjustments. Level 2 for impeller.
How to prevent recurrence	Replace impeller at 200-hour service. Monitor pump pressure continuously during operation.
Warranty impact	N/A — performance optimisation.

SYMPTOM: Discharge distance reduced

Probable causes (most likely first)	1. Worn impeller — reduced head (pressure) capacity. 2. Partial discharge pipe blockage increasing friction loss. 3. Engine not at rated RPM. 4. Pump running at too low a speed.
Diagnostic steps	Check pump pressure at rated RPM — if pressure is below 3 bar, impeller may be worn. Check discharge pipe for partial blockage.
Step-by-step fix / repair	1. Run engine at rated speed and check pressure. 2. Flush discharge pipe with clean water to clear partial blockage. 3. Replace worn impeller (Level 2).
Parts / materials required	Impeller
Tools required	Pressure gauge
Estimated repair time	Flush: 20–30 minutes. Impeller: Level 2.
Operator fix or specialist needed?	Level 1 for flushing and RPM. Level 2 for impeller.
How to prevent recurrence	Replace impeller on schedule. Flush discharge pipe daily at end of shift.
Warranty impact	N/A.

SYMPTOM: Slurry density too low — mostly water being discharged

Probable causes (most likely first)	1. Suction head not close enough to dredging face. 2. Suction too high — pulling surface water rather than bed material. 3. Incorrect anchor positioning — dredger not over undisturbed material.
Diagnostic steps	Observe discharge colour — clear or very light discharge indicates low density. Check suction depth and position.
Step-by-step fix / repair	1. Lower suction arm to bring head closer to seabed/riverbed. 2. Reposition vessel using anchor winches to bring suction over undisturbed material. 3. Increase engine RPM to improve suction velocity.
Parts / materials required	N/A
Tools required	N/A
Estimated repair time	10–20 minutes
Operator fix or specialist needed?	Level 1.
How to prevent recurrence	Monitor discharge colour and density continuously. Reposition as soon as density drops.
Warranty impact	N/A.

7. Preventive Maintenance & Common Customer Queries

Based on the company's experience, the following are the most common customer questions received about these dredgers. These are the questions operators and customers ask most frequently and should be addressed proactively in customer support.

7.1 Most Common Customer Questions

COMMON QUESTION	ANSWER
What kinds of dredgers do you produce?	Four models: 16/14" Cutter Suction Dredger, 14/12" Amphibious Multifunctional Dredger, 12/10" Bucket Chain Dredger, and 10/10" Jet Suction Dredger. Additional models exist in the broader fleet.
What pump sizes do they come in?	16/14 inch, 14/12 inch, 12/10 inch, and 10/10 inch — the first number is suction inlet, the second is discharge outlet diameter.
What distance can it pump / discharge?	16/14": up to 1,000 m. 14/12": up to 600 m. 12/10" and 10/10": confirm with company — depends on pump configuration and soil type.
What size are the engines?	Models 1 & 2: Weichai X6170ZC650-2 (478 kW / 650 HP). Model 3: Caterpillar 3408. Model 4: Weichai WP13 (405 kW / 450 HP). All models have a CAT 3412 auxiliary engine.
What is the dredging depth?	16/14": 22 m. 14/12": 22 m. 10/10": 18 m. Confirm 12/10" depth with company.
What is the size of the pump?	See above — pump size refers to suction/discharge diameter in inches.
What is the maintenance schedule?	Engine oil, fuel filter, hydraulic oil, gearbox oil — all changed every 200 hours. Full overhaul quarterly (90 days). Major engine service annually.
How often do you do maintenance?	Routine checks daily. Oil and filter changes every 200 hours. Quarterly overhaul every 90 days. Full annual service every 12 months.

7.2 Key Spare Parts to Keep on Site

SPARE PART	WHY IT IS CRITICAL
Engine oil — Total Ruby X (10L minimum)	Daily top-up and 200-hour oil change. Running out stops operations immediately.
Fuel filters (primary and secondary)	Replace every 200 hours. Blockage causes engine stalling and power loss.
Air filters (engine)	Replace every 200 hours or when clogged. Clogged filter causes black smoke and power loss.
Gland packing rope (pump seal)	Worn gland causes slurry leakage from pump — replace regularly.
Pump impeller (correct size per model)	Wear item — replace when output drops noticeably. Critical for production.
Anchor wires — correct grade and diameter	Snapped anchor wire stops operations. Always carry one spare length.
Pipe couplings and clamps	Suction and discharge pipe joints wear and leak — keep spares on site.
Battery (correct specification)	Flat battery prevents engine start. Keep one charged spare.
Fuse set (correct ratings)	Blown fuses cause electrical failure. Keep full set of rated fuses.
Gearbox oil (correct grade)	Change every 200 hours — running low causes gearbox overheating and failure.

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Marine Engineering Division

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